

MINOR PROJECT REPORT

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GSN: Graph-Semantic-Net for Vocal Extraction via Harmonic-Aware Refinement

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Declaration

I, **Disha Saini**, Roll No. 23, student of B.Tech 2nd Year, hereby declare that the minor project report titled “*GSN: Graph-Semantic-Net for Vocal Extraction via Harmonic-Aware Refinement*” has been prepared by me under guidance and is submitted as part of the minor project requirement of my programme.

I further declare that this report is my own work and that all sources of information, ideas, and prior work used in this project have been duly acknowledged and cited. This report has not been submitted elsewhere for any other academic award.

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Abstract

This report documents the design, development, and preliminary evaluation of the **GSN (Graph-Semantic-Net)** system, a three-stage hybrid pipeline for vocal extraction from polyphonic music mixtures. The project investigates whether physically-informed and semantically-guided inductive biases can improve separation quality in the specific cases where vocal harmonics and instrumental partials overlap in frequency, a scenario that standard data-driven separators handle poorly.

The pipeline couples a pretrained **Demucs** structural separator, a frozen **CLAP** text-audio embedding model for semantic steering, and a **Complex-Valued Harmonic Graph Convolutional Network (Harmonic GCN)** whose adjacency matrix encodes integer-ratio overtone relationships derived from acoustic physics. The key architectural idea is that the harmonic graph structure is defined from physics and is not learned, giving the network a form of structural prior knowledge that conventional convolutional architectures must discover from data.

Preliminary evaluation across four cumulative configurations shows an improvement in Scale-Invariant Signal-to-Distortion Ratio (SI-SDR) from 3.22 dB for a U-Net baseline to 5.25 dB for the full GSN system. Signal-to-Interference Ratio (SIR) and Signal-to-Artifacts Ratio (SAR) also improve across configurations. The report identifies several important experimental gaps, most notably the absence of a Demucs-alone baseline, and presents a structured plan for closing them in the next phase of work.

Keywords: music source separation, vocal extraction, Demucs, CLAP, graph neural networks, harmonic modeling, complex-valued networks, SI-SDR, inductive bias, physically-informed deep learning.

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I would like to acknowledge the researchers and authors whose prior work forms the foundation of this project. The Demucs architecture developed by Défossez et al. [16], the CLAP model introduced by Elizalde et al. [15], the Graph Convolutional Network formulation of Kipf and Welling [28], the FiLM conditioning mechanism of Perez et al. [32], and the complex-valued network design principles of Trabelsi et al. [?] all provided direct technical foundations for this work.

I also acknowledge the BSS Eval framework by Vincent et al. [34] and the mir_eval library by Raffel et al. [33], which were used for evaluation.

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Chapter 1

Foreword

This report documents the research and development work carried out on the GSN (Graph-Semantic-Net) project during the minor project component of the B.Tech programme. The goal was to investigate whether physically-informed and semantically-guided inductive biases could improve the quality of vocal extraction from mixed audio signals, with a focus on the difficult cases where vocal harmonics and instrumental partials land on the same frequency bins.

The report is written to be factual and measured. It describes what was done, what was borrowed from prior work and from whom, what the results showed, and what remains to be completed. No claims are made beyond what the experiments support, and gaps in the current methodology are named explicitly.

The structure of the report follows a logical progression. Chapter 2 explains why audio engineering matters within the deep learning research landscape. Chapter 3 gives a project overview. Chapter 4 reviews the prior work that the project builds on. Chapter 5 describes what was built. Chapter 6 covers the experimental setup. Chapter 7 presents results. Chapter 8 lists every component taken from prior work. Chapter 9 documents current limitations. Chapter 10 describes what the next phase of work will address. Chapter 11 concludes.

Chapter 2

The Role of Audio Engineering in Deep Learning Research

2.1 Audio as a Distinct Modality

Audio signals carry information very differently from images or text. A single second of audio sampled at 44,100 Hz contains 44,100 individual scalar values. The meaningful content, such as pitch, timbre, rhythm, and phoneme identity, is not stored in individual samples but in the relationships between them across both time and frequency simultaneously [1]. This dual structure makes audio a demanding modality for any machine learning system trying to learn useful representations purely from data.

One important difference from images is the absence of spatial locality as a reliable assumption. In images, nearby pixels tend to be related, which makes convolutional operations a natural architectural fit [2]. In audio, meaningful relationships can span large distances on the frequency axis. As an example, the second harmonic of a note at 220 Hz appears at 440 Hz, which is 220 frequency bins away in a linear frequency representation. A small convolutional kernel would not naturally connect these two bins without being specifically designed to do so or trained on enough examples to discover the relationship [3]. This observation shaped several design decisions in the project.

2.2 Physical Knowledge That Exists Before Any Training

Classical audio engineering has accumulated a detailed understanding of how sound works physically. Some of this knowledge is directly useful for machine learning system design.

The harmonic series is one relevant example. Most pitched sounds, including singing voices, string instruments, wind instruments, and most keyboard instruments, produce energy at integer multiples of a base frequency called the fundamental. This is not a statistical approximation. It follows from the physics of standing waves in vibrating systems [4]. A machine learning model that does not know this must rediscover it from training data, which is inefficient and can make the model brittle in edge cases.

Phase information in the complex frequency representation is another example. The phase

of a spectral coefficient encodes when acoustic events occur in time. When a separator ignores phase and works only on spectral magnitude, it typically damages transients, the sharp note onsets and consonant attacks that make audio sound natural [5]. Audio engineers have understood this for decades, and deep learning systems are only beginning to treat phase preservation as a first-class design concern.

Perceptual loudness and auditory masking form a third area. Human hearing is nonlinear and frequency-dependent. A louder sound can render a softer one in a nearby frequency region effectively inaudible. Psychoacoustic models that account for this effect have been part of audio engineering practice for many years [6] and are increasingly being incorporated into learning-based audio systems.

All three of these knowledge types exist independently of any training dataset. Encoding them into a model architecture means the model does not need to spend representational capacity rediscovering them. This is the core argument for physically-informed design.

2.3 How Deep Learning and Audio Engineering Have Come Together

Deep learning has substantially changed audio research over the past decade. Automatic speech recognition [7], music generation [8], environmental sound classification [9], and music source separation [10] have all advanced through neural approaches. However, purely data-driven methods have a recognizable failure pattern. They perform well on average across a test set while struggling on the specific cases that require structured physical reasoning about sound.

The research community has responded with hybrid approaches that combine learned components with physically-motivated constraints. Examples include incorporating short-time Fourier transform (STFT) representations into waveform models [11], designing filterbanks that resemble the frequency response of the human cochlea rather than using arbitrary learned filters [12], and training with perceptual loss functions that correlate better with human quality judgments than sample-level error measures do [13].

This project sits within that general direction. It does not claim to resolve the broader question of how much physical knowledge should be encoded versus learned. It focuses on one specific instantiation of the idea in the context of vocal extraction from polyphonic music.

Chapter 3

Project Overview

3.1 The Problem

The specific problem addressed in this project is vocal extraction from polyphonic music mixtures: given a recording containing a singing voice mixed with instrumental accompaniment, the task is to recover the vocal signal as cleanly as possible.

This problem has been studied for a long time in the music information retrieval and audio signal processing communities [14]. Practical applications include music production and remixing, karaoke and entertainment systems, accessibility tools for hearing-impaired users, and automatic music content analysis.

The specific difficulty this project focuses on is the harmonic overlap problem. This occurs when a pitched instrument and a singing voice share a pitch class, meaning their harmonic series coincide at the same frequency bins. In those regions, a separator that relies only on spectral energy cannot determine which source owns a given partial without additional information. Standard end-to-end separators learn a statistical association between spectral patterns and source identity. This works in typical cases but produces recognizable artifacts in the hard cases: spectral smoothing, phase errors, and missing or attenuated partials that result in vocals that sound thin or slightly unnatural.

3.2 Approach in Summary

The project builds a three-stage pipeline that addresses the harmonic overlap problem by encoding two types of inductive bias into the system architecture.

The first is a physical bias. An explicit graph structure connects frequency bins that stand in integer-ratio harmonic relationships to each other. Because this structure is defined from acoustic physics rather than learned from data, the model has direct access to harmonic organization without needing to discover it from training examples.

The second is a semantic bias. A frozen text-audio embedding model called CLAP [15]

provides a conceptual description of what vocals are. This allows the refinement stage to be steered by natural language text rather than requiring source-specific label training.

The pipeline uses a pretrained Demucs model [16] as a structural backbone. Demucs handles the bulk of the separation work. The Harmonic Graph Convolutional Network (GCN) refiner and CLAP steering address the residual problem of cleaning up the Demucs output in regions where harmonic ambiguity remains.

3.3 Scope of This Report

This report covers the motivation for each design decision, the implementation of each component, the experimental protocol used, the results obtained, the identified limitations, and the plan for future work. Every component borrowed from prior work is explicitly acknowledged with citations.

Chapter 4

Background and Related Work

4.1 Music Source Separation

Music source separation asks how to decompose a musical mixture into its individual sources such as vocals, bass, drums, and other instruments. It is a specific case of the broader blind source separation problem, which has been studied since the early 1990s [17].

Early methods used tools such as non-negative matrix factorization (NMF) [18] and independent component analysis (ICA) [19]. Non-negative matrix factorization decomposes a spectrogram into a product of basis patterns and activation coefficients that can sometimes be associated with individual sources. Independent component analysis assumes statistical independence between sources, an assumption that is frequently violated in music.

Neural approaches began with deep feedforward networks applied to spectral masking [20]. The idea is to train a network to predict a soft mask in the time-frequency domain that, when multiplied with the mixture spectrogram, suppresses non-target sources. U-Net architectures [21], originally developed for medical image segmentation, were adapted for source separation because their skip connections preserve fine-grained frequency detail across encoding and decoding stages [22].

4.2 Time-Domain Separation and Conv-TasNet

Conv-TasNet [23] reframed the source separation problem by moving from the time-frequency domain to the waveform domain. It uses a learned one-dimensional encoder to map the waveform into a latent representation, a temporal convolutional network to predict source masks in that latent space, and a learned decoder to reconstruct the separated waveforms.

Conv-TasNet showed that learned representations could outperform hand-crafted time-frequency representations on some speech separation benchmarks. It also introduced SI-SDR as a training loss [24], which prevents a model from inflating its score by simply

rescaling the output. We use SI-SDR as the primary evaluation metric in this project, following that convention.

4.3 Demucs

Demucs [16] introduced a waveform-based encoder-decoder architecture for music source separation using a U-Net style structure that operates on raw audio waveforms. It was trained on the MUSDB18 dataset [25] and demonstrated competitive separation quality. A later version called Hybrid Demucs [11] combined waveform and spectrogram processing branches to improve performance further.

The pretrained Demucs model is used as the backbone of the pipeline in this project. Its weights are not modified during training of the refinement stages. Freezing Demucs is a practical choice: it provides a strong structural estimate that is more useful as a starting point than training a new separator from scratch with limited resources.

4.4 CLAP: Contrastive Language-Audio Pretraining

CLAP [15] is a model that learns aligned representations of audio and natural language text using contrastive learning, following the design pattern established by CLIP [26] for vision and language. A CLAP model trained on large audio-text datasets can retrieve audio from text queries and perform audio classification in a zero-shot manner without task-specific training data.

CLAP is used in this project as a source of semantic conditioning rather than as a separator. The reasoning is that a frozen CLAP model has already learned what “lead vocals” means as a concept in a shared audio-text embedding space. Providing that concept embedding to the refinement stage gives it semantic guidance that purely signal-level approaches cannot offer. The CLAP model is used exactly as released by its authors [15] with no fine-tuning.

4.5 Graph Neural Networks

Graph neural networks [27] generalize convolution to data structured as graphs rather than grids. A standard graph convolutional network layer [28] aggregates node features from neighbors using a normalized adjacency matrix, allowing information to flow along edges that encode meaningful relationships between nodes.

Graph neural networks have been applied to audio and music tasks including music transcription [29] and chord recognition [30]. The application in this project is different:

the graph is defined in the frequency domain of a spectrogram, where nodes are frequency bins and edges connect bins standing in integer-ratio harmonic relationships. The graph structure is defined by physical acoustics and is not learned from data, which is the key design choice distinguishing this approach.

4.6 Complex-Valued Neural Networks

Standard neural networks operate on real-valued numbers. Complex-valued neural networks [31] extend this to complex numbers, allowing a network to process complex spectral representations without separating magnitude and phase into unrelated pathways.

Retaining phase information is important for reconstructing transients accurately [5]. When a separator operates only on spectral magnitude and relies on the mixture phase for reconstruction, errors in the phase relationship between components produce smeared transients and artifacts at note onsets. In this project, a complex STFT of the Demucs output is computed, and complex-valued node features are passed into the Harmonic GCN so that phase information is preserved through the refinement step.

4.7 FiLM Conditioning

Feature-wise Linear Modulation, abbreviated as FiLM [32], is a conditioning technique that allows an external input to modulate the internal feature representations of a neural network. It works by predicting per-feature scale and shift parameters that are applied at intermediate network layers. FiLM was originally developed for visual question answering and has since been used broadly in conditional generative modeling.

FiLM-style conditioning is used in this project to inject the CLAP semantic embedding into the Harmonic GCN layers. The CLAP embedding is projected through a linear layer to produce per-feature scale and shift vectors that are broadcast across frequency bin nodes at each graph convolutional layer.

4.8 Evaluation Metrics

Three standard metrics from the source separation community are used, all computed using the `mir_eval` library [33]. The metric definitions originate from the BSS Eval framework introduced by Vincent et al. [34].

SI-SDR (Scale-Invariant Signal-to-Distortion Ratio) [24] is the primary metric. It measures separation quality while being invariant to output scaling.

SIR (Signal-to-Interference Ratio) measures how well energy from non-target sources is suppressed in the output.

SAR (Signal-to-Artifacts Ratio) measures how much processing artifact the separator introduces into the output.

Using all three together provides a more complete picture than any single metric alone. A system that aggressively suppresses interference shows high SIR but low SAR; a system that is conservative about masking shows the opposite pattern. The combination makes these trade-offs visible.

Chapter 5

What We Built: System Description

5.1 Pipeline Overview

The GSN pipeline has three stages operating in sequence.

Stage 1 is structural separation using a frozen Demucs model. It produces an initial vocal estimate from the mixture signal.

Stage 2 is semantic steering using a frozen CLAP model. It computes a conditioning vector from a natural language text prompt.

Stage 3 is harmonic refinement using a Complex-Valued Harmonic Graph Convolutional Network. It refines the Demucs estimate by propagating evidence along physically-defined harmonic edges in a frequency-domain graph, conditioned on the CLAP embedding.

Only Stage 3 contains trainable parameters. Stages 1 and 2 are frozen throughout training and inference.

5.2 Stage 1: Structural Separation

Given a mixture signal $x(t)$, it is passed through a frozen pretrained Demucs model [16] to obtain an initial vocal estimate:

$$\hat{v}_0(t) = f_{\text{Demucs}}(x(t)) \quad (5.1)$$

Demucs handles the broad-strokes work of separating vocals from drums, bass, and pitched accompaniment. It performs well in most frequency regions. The remaining errors tend to concentrate in regions of harmonic overlap, which the subsequent stages address.

5.3 Stage 2: Semantic Steering

Two text prompts are defined. One describes the target: “*lead vocals, singing voice.*” The other describes what should be excluded: “*instrumental accompaniment, guitar, synth.*”

Using the frozen CLAP text encoder [15], an embedding is computed for each prompt:

$$e_{\text{voc}} = \text{CLAP}_{\text{text}}(p_{\text{voc}}), \quad e_{\text{bleed}} = \text{CLAP}_{\text{text}}(p_{\text{bleed}}) \quad (5.2)$$

An audio embedding of the Demucs output is also computed:

$$e_a = \text{CLAP}_{\text{audio}}(\hat{v}_0) \quad (5.3)$$

A semantic steering scalar α is then derived from the cosine similarities of the audio embedding to each prompt embedding:

$$\alpha = \sigma\left(\tau[\cos(e_a, e_{\text{voc}}) - \cos(e_a, e_{\text{bleed}})]\right) \quad (5.4)$$

where σ denotes the sigmoid function and τ is a temperature parameter. This scalar reflects how closely the current Demucs output resembles the vocal concept versus the instrumental concept in CLAP’s shared embedding space.

The vocal prompt embedding is also projected through a linear layer to produce a conditioning vector $\mathbf{c}$ that is passed to the GCN at each layer using the FiLM mechanism [32].

5.4 Stage 3: Harmonic Graph Convolutional Network

5.4.1 Spectral Input Representation

The complex STFT of the Demucs output $\hat{v}_0(t)$ is computed to give $V_0(f, n) \in \mathbb{C}$. For each time frame n , a graph is defined where nodes correspond to frequency bins. The feature vector at each node f is:

$$\mathbf{h}_f^{(0)} = [\Re(V_0(f, n)), \Im(V_0(f, n)), |V_0(f, n)|]^\top \quad (5.5)$$

Retaining real and imaginary parts separately preserves phase information through the processing chain [31].

5.4.2 The Harmonic Adjacency Matrix

Let $F(f)$ denote the center frequency of bin f . For maximum harmonic order K and frequency tolerance ε , the adjacency matrix A is defined as:

$$A_{ij} = \begin{cases} w_k \cdot \exp\left(-\frac{|F(f_j) - kF(f_i)|}{\sigma_f}\right), & \exists k \in \{2, \dots, K\} : |F(f_j) - kF(f_i)| \leq \varepsilon \\ 0, & \text{otherwise} \end{cases} \quad (5.6)$$

where w_k is a weight that decreases with harmonic order k (higher harmonics are less stable and more easily masked by interference), and σ_f controls the bandwidth of the Gaussian weighting to accommodate vibrato and gradual pitch drift.

Symmetric normalization following the standard GCN convention [28] is applied:

$$\tilde{A} = D^{-\frac{1}{2}} (A + I) D^{-\frac{1}{2}} \quad (5.7)$$

where D is the degree matrix. The adjacency matrix is computed from physical acoustics before training begins and is not updated during training. Only the message-passing weight matrices within the GCN layers are learned.

5.4.3 Message Passing and Output

Each graph convolutional layer computes:

$$H^{(\ell+1)} = \phi\left(\tilde{A}H^{(\ell)}W^{(\ell)} + \gamma(\mathbf{c})\right) \quad (5.8)$$

where ϕ is a nonlinear activation function, $W^{(\ell)}$ is the learnable weight matrix at layer ℓ , and $\gamma(\mathbf{c})$ is the FiLM conditioning term [32] derived from the CLAP semantic embedding.

The final layer produces a complex-valued residual ΔV in the frequency domain. The refined spectrogram is:

$$V_{\text{ref}} = V_0 + \alpha \cdot \Delta V \quad (5.9)$$

The reconstructed waveform is obtained via the inverse STFT:

$$\hat{v}(t) = \text{iSTFT}(V_{\text{ref}}) \quad (5.10)$$

5.4.4 Training Objective

The Harmonic GCN is trained by minimizing a composite loss with four terms:

- (1) Negative SI-SDR [24] between the reconstructed waveform $\hat{v}(t)$ and the ground-truth vocal signal.
- (2) A complex STFT magnitude loss between the refined and reference spectrograms.
- (3) A phase-consistency term encouraging agreement between the refined and reference phase.
- (4) A harmonic regularization term encouraging the energy at harmonically-connected bins to evolve coherently across time frames.

The relative weighting of these four terms requires careful tuning. If the harmonic regularizer is weighted too strongly, it can over-smooth harmonic structure and damage transients, which is the opposite of the intended effect.

5.5 Inference Design

The three components of the pipeline have different computational costs and different rates of change during inference. The CLAP text encoder runs once per text prompt, since a user’s intent does not change within a session. The CLAP audio encoder can be run sparsely, roughly every few seconds, because the semantic character of audio changes slowly relative to the frame rate. Demucs and the Harmonic GCN run once per audio frame.

Because the Harmonic GCN operates on a sparse adjacency matrix where only harmonically-related bins are connected, its per-frame cost is substantially lower than a dense attention mechanism. On the reference hardware used in this project, this design yields a Real-Time Factor (RTF) below 0.30, meaning the system processes audio faster than it is produced in real time. This figure applies to the specific hardware configuration tested and would need to be measured separately on other platforms.

Chapter 6

Experimental Work

6.1 System Configurations Tested

Four cumulative configurations were evaluated to understand the contribution of each component.

- Phase 1** U-Net baseline: a standard spectral masking network with no refinement. This is the starting reference point.
- Phase 2** U-Net baseline with Harmonic GCN refinement added, without any CLAP conditioning.
- Phase 3** Phase 2 system with CLAP semantic steering added.
- Phase 4** The full GSN system: Demucs backbone combined with Harmonic GCN and CLAP steering.

An important caveat applies to the comparison between Phase 3 and Phase 4. The backbone changes at that transition: Phases 1 through 3 use a U-Net base, while Phase 4 uses Demucs. This means the Phase 4 result reflects both the improved backbone and the refinement modules combined. Cleanly isolating the contribution of the novel modules requires a Demucs-alone baseline, which has not yet been run. This is identified as a gap in Section 6.3.

6.2 Evaluation Metrics

SI-SDR is the primary metric [24]. SIR and SAR are reported alongside it. All three metrics are computed using the `mir_eval` library [33], following the BSS Eval conventions introduced by Vincent et al. [34].

6.3 Current Gaps in the Experimental Setup

The following gaps are identified openly.

Missing dataset specification. The evaluation was conducted on a dataset that has not been formally documented in this report. Its corpus identity, size, genre coverage, and train-test split have not been described. Without this, the results cannot be compared to published benchmarks or reproduced by others.

No statistical testing. All results are single-point estimates with no confidence intervals and no variance across repeated runs. No statistical significance testing has been performed. Small differences in the table should therefore be interpreted with caution.

Missing Demucs-alone baseline. This is the most important gap. Without knowing the SI-SDR of Demucs applied to the same data without any refinement, the contribution of the Harmonic GCN and CLAP modules to the Phase 4 result cannot be isolated.

Chapter 7

Results

7.1 Quantitative Results

Table 7.1 reports SI-SDR, SIR, and SAR for each evaluated configuration. Higher values are better for all three metrics.

Table 7.1: Evaluation results across four system configurations. All values are in decibels (dB). Higher is better. Results are preliminary single-point estimates.

Configuration	SI-SDR (dB)	SIR (dB)	SAR (dB)
U-Net Baseline (Phase 1)	3.22	5.10	4.80
Plus Harmonic GCN (Phase 2)	3.12	5.35	4.10
Plus CLAP Steering (Phase 3)	3.80	6.02	5.20
Full GSN System (Phase 4)	5.25	8.15	7.40

7.2 Observations

Phase 2. SI-SDR decreases slightly from 3.22 to 3.12 dB, while SIR increases from 5.10 to 5.35 dB and SAR falls from 4.80 to 4.10 dB. One plausible reading of this pattern is that the graph structure improves interference suppression (shown by rising SIR) while introducing reconstruction artifacts (shown by falling SAR) because the GCN has not yet received semantic guidance about which harmonics to preserve. This interpretation has not been confirmed by frequency-resolved analysis and should be treated as a hypothesis.

Phase 3. Adding CLAP conditioning brings SI-SDR to 3.80 dB, SIR to 6.02 dB, and SAR to 5.20 dB. The improvement in SAR suggests that the semantic embedding helps the GCN make better reconstruction decisions in harmonically ambiguous regions.

Phase 4. All three metrics improve substantially: SI-SDR to 5.25 dB, SIR to 8.15 dB, and

SAR to 7.40 dB. As noted in Chapter 6, this improvement includes the effect of switching the backbone from U-Net to Demucs. The contribution of each component within Phase 4 has not been isolated.

7.3 What the Results Do Not Show

The current results do not show whether the gains are statistically significant. They do not show how GSN compares to Demucs without refinement. They do not show how these numbers relate to published benchmarks because the evaluation dataset is unspecified. These are open questions that future experiments need to address before the results can be treated as confirmed findings.

Chapter 8

What We Took From Others

This chapter explicitly lists every component and technique that originated in prior work. Being transparent about which parts of the system are novel and which are adopted from existing work is an important part of honest reporting.

Table 8.1: Components adopted from prior work with corresponding references.

Component	Source	Reference
Demucs backbone model	Défossez, Usunier, Bottou, Bach	[16]
Hybrid Demucs variant	Défossez	[11]
CLAP text and audio encoders	Elizalde, Deshmukh, Al Ismail, Wang	[15]
Contrastive text-audio pre-training concept	Radford et al. (CLIP)	[26]
SI-SDR metric and training loss	Luo and Mesgarani; Le Roux et al.	[23, 24]
GCN layer formulation and normalization	Kipf and Welling	[28]
Complex-valued network design	Trabelsi et al.	[31]
FiLM conditioning mechanism	Perez et al.	[32]
U-Net architecture (used as Phase 1 baseline)	Ronneberger, Fischer, Brox	[21]
mir_eval evaluation library	Raffel et al.	[33]
BSS Eval metric framework	Vincent et al.	[34]
MUSDB18 evaluation dataset (if confirmed)	Raffi et al.	[25]

The components developed within this project are the harmonic adjacency matrix construction from integer-ratio frequency relationships, the three-stage pipeline design coupling these frozen pretrained models, and the adaptation of the FiLM conditioning mechanism to operate within graph convolutional layers. Prior work has applied graph neural networks to music tasks [29, 30], but the use of a physically-defined harmonic adjacency as the fixed structural basis for a GCN refiner in a source separation pipeline is, to the best of our knowledge, novel.

Chapter 9

Current Limitations

This chapter documents the limitations of the current system and experimental protocol. These are not minor caveats. They affect how much confidence can be placed in the results in their current form.

9.1 Missing Demucs Baseline

The most important missing piece is a Demucs-alone result on the same evaluation data. The Phase 4 result of 5.25 dB SI-SDR cannot be attributed primarily to the Harmonic GCN or CLAP modules without knowing what Demucs achieves without them on the same test set. Until this baseline is produced, the attribution of performance to the novel components remains unclear.

9.2 Unspecified Evaluation Dataset

The evaluation dataset has not been formally described. Until it is, the numbers in Table 7.1 cannot be compared to any published benchmark and cannot be reproduced by an outside reader. This is a significant limitation for any external validity claim.

9.3 No Statistical Testing

Every number in Table 7.1 is a single estimate from a single evaluation run. A difference of 0.10 dB between Phase 1 and Phase 2 SI-SDR could easily fall within the noise of the evaluation procedure. Strong claims about small differences require repeated runs and formal significance testing, neither of which have been conducted.

9.4 Unpitched Vocal Components

The harmonic graph assumes that the target source is harmonically structured. Unpitched vocal content, including breath sounds, fricative consonants such as /s/, /f/, and //, and

plosive transients, does not follow the integer-ratio pattern. If the GCN aggressively suppresses non-harmonic content, it may unintentionally remove these components from the output. The extent of this effect has not been measured.

9.5 CLAP Prompt Sensitivity

The behavior of the semantic steering stage depends on the specific wording of the text prompts. No systematic test of different prompt formulations has been conducted. Some prompt choices may produce degraded results, particularly if the wording maps to regions of CLAP’s embedding space that overlap with instrument timbres.

9.6 Harmonic Regularizer Sensitivity

The harmonic regularization term used during training affects how strongly the GCN enforces temporal coherence in the harmonic structure. The sensitivity of the final SI-SDR to the weighting of this term relative to the other loss components has not been characterized.

9.7 Hardware-Specific RTF Measurement

The Real-Time Factor below 0.30 was measured on one hardware configuration. Performance on other platforms, including CPU-only machines or mobile hardware, is unknown and would need to be measured separately.

Chapter 10

What We Are Working on Next

10.1 Immediate Priority: Closing the Experimental Gaps

The first and most urgent task is running the frozen Demucs model alone on the evaluation data and adding that result to the comparison table. This single experiment will clarify how much of the Phase 4 improvement comes from the backbone versus the refinement modules.

The second task is formally documenting the evaluation dataset, including its name, size, genre distribution, and the construction of train, validation, and test splits.

The third task is repeating the full evaluation across at least three random seeds and reporting mean and standard deviation for each configuration. A paired statistical significance test will be applied to the headline SI-SDR comparison.

The fourth task is building a cleaner ablation design. A two-by-two table varying backbone (U-Net versus Demucs) and refinement modules (none versus GCN plus CLAP) independently will separate the contributions of backbone quality and refinement module design in a way the current table cannot.

10.2 Near-Term Technical Analysis

After the experimental gaps are closed, a set of deeper analytical experiments will follow.

A frequency-resolved decomposition of the SIR and SAR results will identify which frequency bands benefit from the harmonic graph and which do not. This will either confirm or refute the hypothesis proposed in Section 7.2 about the Phase 2 result pattern.

A sensitivity analysis of the harmonic regularizer weight will characterize how much the final SI-SDR changes as this term is varied, which will inform training decisions for future configurations.

A systematic prompt sensitivity study with the CLAP steering module will identify which

prompt formulations are robust and which lead to degraded results.

Fricative and breath-noise segments will be isolated in the test data to measure directly whether the Harmonic GCN suppresses them relative to the no-refinement condition.

10.3 Longer-Term Development

After the analysis work is complete, the following directions are planned.

A fully causal streaming version of the pipeline is needed for true real-time online operation. This requires modifying the STFT to use causal windowing and redesigning the graph message-passing so it aggregates only from past frames.

Quantization-aware training and graph sparsification could make the system practical on mobile or embedded hardware. The Harmonic GCN is already compact, but Demucs is not, and reducing its inference cost on device would require additional engineering work.

A small-scale listening study using MUSHRA methodology [35] is planned to assess whether the metric improvements correspond to perceptible quality differences for human listeners. SI-SDR is a useful proxy for separation quality but is not equivalent to perceived quality.

Finally, the harmonic graph approach will be tested on sources other than vocals. Bass guitar, piano, and most wind instruments are harmonically structured in the same way that singing voices are. The adjacency matrix construction would apply without fundamental modification, and it is informative to know whether the refinement approach generalizes across source types.

Chapter 11

Conclusion

This report has described the GSN project from its motivation through its implementation to its preliminary results and remaining gaps.

The central problem is that harmonic overlap between vocals and instruments creates ambiguous regions in the frequency representation that statistical separators handle inconsistently. The central idea is that encoding integer-ratio harmonic relationships as a fixed graph structure gives a GCN refiner a form of physical prior knowledge that it would otherwise need to learn from data, and that pairing this with a frozen text-audio embedding model provides semantic guidance that purely signal-level approaches lack.

The preliminary results show improvement across all three metrics from the U-Net baseline to the full system. The full system reaches 5.25 dB SI-SDR compared to 3.22 dB for the U-Net baseline, with accompanying gains in SIR from 5.10 dB to 8.15 dB and in SAR from 4.80 dB to 7.40 dB.

However, significant experimental gaps remain. The most important is the absence of a Demucs-alone baseline, which means the contribution of the novel components to the Phase 4 result has not been isolated. Until the gaps described in Chapters 6 and 9 are addressed, the results should be treated as preliminary indicators rather than confirmed findings.

The project is ongoing. The next phase of work is focused on closing the experimental gaps before any further development of the system architecture.

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